Processes and page tables: activity

What does the operating system do so many programs can share one machine?

IT 520 · November 2 · 20 minutes · Work in pairs

Name(s): _____

The same laptop as in class: one CPU running a script, a browser and a shell. Show your steps.

Part 1. Taking turns (about 7 minutes)

> **Rules for Part 1**
> Turns go Browser, Script, Shell, then repeat. Each turn is a time slice of 2 ms.
> The next turn goes to the next process in that order after the one that just ran. Skip Blocked and Exited processes.
> A finished read makes a process Ready. It does not take the CPU from the process that is running.
> There is one CPU, so exactly one process is Running at a time.

1. Here are the events, in order.

0 ms: The CPU starts the browser.
1 ms: The browser starts a disk read, to load a saved image.
3 ms: The script's slice ends.
4 ms: The shell ends, because you typed `exit`.
5 ms: The browser's disk read finishes.
6 ms: The script's slice ends.

(a) Fill in each process's state just after each event. Use Running, Ready, Blocked or Exited. The first row is done for you.

Time	Who is Running?	Browser	Script	Shell
0 ms	Browser	Running	Ready	Ready
1 ms				
3 ms				
4 ms				
5 ms				
6 ms				

(b) The browser's read is done at 5 ms. Why does the browser not get the CPU until 6 ms?

2. The operating system ran at 1 ms and again at 5 ms.

(a) For each time, say what made it run: a system call from a program, or an interrupt from a device. Name the program or the device.

(b) The browser runs in user mode. Why does it ask the operating system for the read instead of reading the disk itself?

Part 2. Private addresses (about 5 minutes)

Rules for Part 2

Pages are 4 KB (0x1000 bytes), so the last three hex digits of an address are the offset.

The frame numbers below are examples.

Page	Script's page table	Browser's page table
0x210	frame 0x3A5	frame 0x4C8
0x211	not in RAM	frame 0x51E
0x212	frame 0x3A6	frame 0x2D0

3. The script and the browser both use the virtual address 0x210B7C.

(a) Find the physical address for each one. Show the split into page and offset.

(b) A classmate says the script can reach the browser's data by using the browser's physical address from (a) as its own address. Is that right? Use the script's page table.

4. A classmate translated the browser's virtual address 0x211004 and wrote “not in RAM, page fault.” What went wrong? Give the right physical address and show the split.

5. The script uses the virtual address 0x211F00. Page 0x211 is a legal page of the script's. It is stored on disk, and a free frame is available. The browser is Ready.

(a) Can you write a physical address for 0x211F00 right now? Why or why not?

(b) What does the operating system do next? Which state is the script in while it waits?

(c) A classmate says that once the page is loaded, the script goes straight back to Running. Is that right? Why or why not?

For the debrief, no need to write: can the browser use the CPU while the script waits?

Before you leave

6. Finish the sentence: Ready differs from Blocked because …